

This assignment will not be graded and is only for practice.

5.1 Level 1:

1) Choose the set of correct options

1 point

- ☐ If A is a real 3×3 matrix, then $\det(A) = \det(A^T)$.
- ☐ If $A = [a_{ij}]$ is a real 4×4 matrix, then the order of the sub matrix obtained by deleting the i -th row and j -th column of A is 4×4 .
- ☐ If $A = [a_{ij}]$ is a real 4×4 matrix, then the order of the sub matrix obtained by deleting the i -th row and j -th column of A is 3×3 .
- ☐ If A is a real 3×3 matrix, then the orders of all possible square sub matrices of A are 1×1 , 2×2 , and 3×3 .

No, the answer is incorrect.

Score: 0

Accepted Answers:

If A is a real 3×3 matrix, then $\det(A) = \det(A^T)$.

If $A = [a_{ij}]$ is a real 4×4 matrix, then the order of the sub matrix obtained by deleting the i -th row and j -th column of A is 3×3 .

If A is a real 3×3 matrix, then the orders of all possible square sub matrices of A are 1×1 , 2×2 , and 3×3 .

2) Let $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & 2 \\ 2 & 1 & 1 \end{bmatrix}$ be a 3×3 matrix. Which of the following is(are) correct?

1 point

[Hint: Determinant can be calculated by expanding with respect to different rows or columns.]

- ☐ $\det(A) = 3 \times \det \begin{pmatrix} 3 & 2 \\ 2 & 1 \end{pmatrix} - 2 \times \det \begin{pmatrix} 2 & 2 \\ 2 & 1 \end{pmatrix} + 2 \times \det \begin{pmatrix} 2 & 3 \\ 2 & 1 \end{pmatrix}$
- ☐ $\det(A) = 3 \times \det \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix} + 2 \times \det \begin{pmatrix} 2 & 2 \\ 1 & 2 \end{pmatrix} + 2 \times \det \begin{pmatrix} 2 & 3 \\ 2 & 1 \end{pmatrix}$
- ☐ $\det(A) = -2 \times \det \begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix} + 3 \times \det \begin{pmatrix} 3 & 2 \\ 2 & 1 \end{pmatrix} - 2 \times \det \begin{pmatrix} 3 & 2 \\ 2 & 1 \end{pmatrix}$

☐ $\det(A) = 2 \times \det\left(\begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}\right) - 3 \times \det\left(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}\right) + 2 \times \det\left(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}\right)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\det(A) = 3 \times \det\left(\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}\right) - 2 \times \det\left(\begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}\right) + 2 \times \det\left(\begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix}\right)$$

$$\det(A) = 3 \times \det\left(\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}\right) + 2 \times \det\left(\begin{bmatrix} 2 & 2 \\ 1 & 2 \end{bmatrix}\right) + 2 \times \det\left(\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}\right)$$

$$\det(A) = -2 \times \det\left(\begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}\right) + 3 \times \det\left(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}\right) - 2 \times \det\left(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}\right)$$

3) If all the elements of a 3×3 real matrix A are the same, then which of the following is (are) **1 point** correct?

- ☐ Determinant of matrix A is 0.
- ☐ Determinant of matrix A cannot be determined from the given information.
- ☐ Determinant of matrix A will be the sum of the elements of a row.
- ☐ Determinant of matrix $A + A^T$ is 0, where A^T denotes the transpose of A .
- ☐ Determinant of matrix $A + A^T$ cannot be determined from the given information, where A^T denotes the transpose of A .

No, the answer is incorrect.

Score: 0

Accepted Answers:

Determinant of matrix A is 0.

Determinant of matrix $A + A^T$ is 0, where A^T denotes the transpose of A .

4) Choose the set of correct options

1 point

- ☐ If two rows are the same in a 3×3 real matrix, then the determinant of that matrix is zero.
- ☐ If two columns are the same in a 3×3 real matrix, then the determinant of that matrix is zero.

- ☐ If one row is a non-zero multiple of another row in a 3×3 real matrix, then the determinant of that matrix is not zero.
- ☐ If one column is a non-zero multiple of another column in a 3×3 real matrix, then the determinant of that matrix is zero.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If two rows are the same in a 3×3 real matrix, then the determinant of that matrix is zero.

If two columns are the same in a 3×3 real matrix, then the determinant of that matrix is zero.

If one column is a non-zero multiple of another column in a 3×3 real matrix, then the determinant of that matrix is zero.

5) Let $A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ be a matrix of order 4×4 . Find $\det(A)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -1

1 point

5.2 Level 2:

6) Let A , B , and C be 3×3 real matrices. Which of the following options is (are) correct? **1 point**

- ☐ $\det(ABC) = \det(A)\det(B)\det(C)$
- ☐ $\det(A^3) = \det(A)^3$
- ☐ $\det(A + B + C) = \det(A) + \det(B) + \det(C)$
- ☐ $\det(AB^T) = \det(A) \det(B)$, where B^T denotes the transpose of B .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\det(ABC) = \det(A)\det(B)\det(C)$$

$$\det(A^3) = \det(A)^3$$

$$\det(AB^T) = \det(A) \det(B), \text{ where } B^T \text{ denotes the transpose of } B.$$

7) Let $A = [\delta_{ij}]$ be a matrix of order 3×3 such that $\delta_{ij} = \begin{cases} 0 & \text{if } i > j \\ j & \text{if } i \leq j \end{cases}$

1 point

Choose the set of correct options

[Hint: Write down matrix A first.]

- ☐ A is an upper triangular matrix.
- ☐ A is a lower triangular matrix.
- ☐ $\det(A) = 6$
- ☐ $\det(A) = 3$

No, the answer is incorrect.

Score: 0

Accepted Answers:

A is an upper triangular matrix.

$$\det(A) = 6$$

8) Let $A = [\delta_{ij}]$ be a matrix of order 3×3 such that $\delta_{ij} = \begin{cases} 0 & \text{if } i < j \\ 2 & \text{if } i \geq j \end{cases}$

1 point

Choose the set of correct options

- ☐ A is an upper triangular matrix.
- ☐ A is a lower triangular matrix.
- ☐ $\det(A) = 2$
- ☐ $\det(A) = 8$

No, the answer is incorrect.

Score: 0

Accepted Answers:

A is a lower triangular matrix.

$\det(A) = 8$

9) Suppose $A = \begin{bmatrix} 4 & -1 & 3 \\ 2 & 0 & 1 \\ 3 & -2 & 0 \end{bmatrix}$ and C_{ij} is the (i, j) -th cofactor of the matrix A .

Let B be 3×3 matrix where (i, j) -th entry is equal to C_{ij} . Find $\det(B)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 49

1 point

10) Suppose A and B are two 3×3 matrices such that $\det(A) = 4$ and $B = 3A$. Find the value of $\sqrt[3]{\det(A^2 B)}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 12

1 point

Your score is: 0/10